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[21AS731]

SEVENTH SEMESTER B.Tech. EXAMINATION, DECEMBER – 2024

AEROSPACE VEHICLE DESIGN

Time:3 Hours **Maximum Marks:50**

Instructions:
- i. PART - A (Question 1-5) answer any 4 questions.
- ii. PART - B (Question 6-8) answer any 2 questions.
- iii. PART - C (Question 9) is compulsory

		PART – A			**05X04=20**
1.		Plot lift distribution in the chord and span wise directions for a simple wing plan form and compare with delta wing configuration.	L2	CO2	05
2.		Draw stress-strain curves for metal and composites on a single plot. Compare strength criteria.	L3	CO2	05
3.		With neat diagram describe the procedure to calculate overall C.G. of the aircraft from various subsystem masses.	L3	CO2	05
4.		Define equilibrium and stability. Explain stable, unstable and Critical equilibrium for a Solid long Column subjected to axial compressive load.	L3	CO2	05
5.		Explain aerodynamically stable, unstable configuration in aircraft Design based on A_C (Aerodynamic center) and C_P (Centre of Pressure) location in chord wise direction. In which case unstable it is advantageous?	L3	CO3	05
		PART - B			**09X02=18**
6.	a.	Suggest a concept design with performance, cabin seating layout, specifications and subsystem selection for a 60-seat passenger twin-engine aircraft (including pilots and two cabin crew members) for operating range of 2000km.	L3	CO3	05
	b.	For the question 6a aircraft draw a normal V-n Diagram.	L3	CO3	04
7.	a.	For the question 6a aircraft describe power plant selection criteria at Max (L/D) condition, suggest a reliable engine available in market for this.	L3	CO3	05
	b.	Draw engine cross section of the above selected engine.	L4	CO3	
8.	a.	With a neat diagram briefly discuss rocket stages concept design for a 6.5 Ton satellite payload class for Geo Synchronous orbit. Give an example.	L4	CO4	05

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	b.	Describe satellite application areas depending its Orbit. In this regard, explain ASLV, PSLV, GSLV-Mk2, GSLV Mk-3.	L4	CO4	04
		PART - C			**01X12=12**
9.	a.	Briefly discuss latest technology practices in aircraft industry for design, analysis and manufacturing using CAD/CAE/PLM, give two examples of each from commercial software	L4	CO4	05
	b.	Discuss briefly basic specifications, unique features of flowing aircrafts: (I) Passenger class: Boeing 787, Airbus 380. (ii) Military class: Rafael, LCA-Mk1, Mk2 and AMCA	L4	CO4	04